

Declarations

CRedit authorship contribution statement. *[AUTHOR: e.g. Name: Conceptualization, Methodology, Software, Investigation, Writing – original draft. Name: Supervision, Writing – review & editing.]*

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding. *[AUTHOR: Funding source and grant number, or “This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.”]*

Data availability. IDRiD and APTOS-2019 are publicly available from their providers [12, 24]. The test-set indices, per-image predictions and all metrics are produced by the released code.

Code availability. *[AUTHOR: Repository URL (e.g. GitHub + Zenodo DOI) containing the notebook, export cell and trained weights.]*

Ethics statement. This study used only publicly available, de-identified datasets; no new data were collected from human participants.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to help draft and structure the manuscript text from the authors’ code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. *[AUTHOR: Keep this statement accurate; most publishers, including Elsevier, require it.]*

Acknowledgements. *[AUTHOR: Optional.]*

References

- [1] Joanne W. Y. Yau, Sophie L. Rogers, Ryo Kawasaki, Ecosse L. Lamoureux, Jonathan W. Kowalski, Toke Bek, Shih-Jen Chen, Jacqueline M. Dekker, Astrid Fletcher, Jakob Grauslund, et al. Global prevalence and major risk factors of diabetic retinopathy. *Diabetes Care*, 35(3):556–564, 2012. doi: 10.2337/dc11-1909.
- [2] Zhen Ling Teo, Yih-Chung Tham, Marco Yu, Miao Li Chee, Tyler Hyungtaek Rim, Ning Cheung, Mukharrem M. Bikbov, Ya Xing Wang, Yating Tang, Yi Lu, et al. Global prevalence of diabetic retinopathy and projection of burden through 2045: Systematic review and meta-analysis. *Ophthalmology*, 128(11):1580–1591, 2021. doi: 10.1016/j.ophtha.2021.04.027.
- [3] Daniel Shu Wei Ting, Gemmy Chui Ming Cheung, and Tien Yin Wong. Diabetic retinopathy: global prevalence, major risk factors, screening practices and public health challenges: a review. *Clinical & Experimental Ophthalmology*, 44(4):260–277, 2016. doi: 10.1111/ceo.12696.
- [4] C. P. Wilkinson, Frederick L. Ferris, Ronald E. Klein, Paul P. Lee, Carl David Agardh, Matthew Davis, Diana Dills, Anselm Kampik, R. Pararajasegaram, and Juan T. Verdager. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. *Ophthalmology*, 110(9):1677–1682, 2003. doi: 10.1016/S0161-6420(03)00475-5.

- [5] Jonathan Krause, Varun Gulshan, Ehsan Rahimy, Peter Karth, Kasumi Widner, Greg S. Corrado, Lily Peng, and Dale R. Webster. Grader variability and the importance of reference standards for evaluating machine learning models for diabetic retinopathy. *Ophthalmology*, 125(8):1264–1272, 2018. doi: 10.1016/j.ophttha.2018.01.034.
- [6] Varun Gulshan, Lily Peng, Marc Coram, Martin C. Stumpe, Derek Wu, Arunachalam Narayanaswamy, Subhashini Venugopalan, Kasumi Widner, Tom Madams, Jorge Cuadros, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA*, 316(22):2402–2410, 2016. doi: 10.1001/jama.2016.17216.
- [7] Rishab Gargeya and Theodore Leng. Automated identification of diabetic retinopathy using deep learning. *Ophthalmology*, 124(7):962–969, 2017. doi: 10.1016/j.ophttha.2017.02.008.
- [8] Daniel Shu Wei Ting, Carol Yim-Lui Cheung, Gilbert Lim, Gavin Siew Wei Tan, Nguyen D. Quang, Alfred Gan, Haslina Hamzah, Renata Garcia-Franco, Ian Yew San Yeo, Shu Yen Lee, et al. Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *JAMA*, 318(22):2211–2223, 2017. doi: 10.1001/jama.2017.18152.
- [9] Ling Dai, Liang Wu, Huating Li, Chun Cai, Qiang Wu, Hongyu Kong, Ruhan Liu, Xiangning Wang, Xuhong Hou, Yuexing Liu, et al. A deep learning system for detecting diabetic retinopathy across the disease spectrum. *Nature Communications*, 12:3242, 2021. doi: 10.1038/s41467-021-23458-5.
- [10] Michael D. Abràmoff, Philip T. Lavin, Michele Birch, Nilay Shah, and James C. Folk. Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *npj Digital Medicine*, 1:39, 2018. doi: 10.1038/s41746-018-0040-6.
- [11] Xiaoxuan Liu, Livia Faes, Aditya U. Kale, Siegfried K. Wagner, Dun Jack Fu, Alice Bruynseels, Thushika Mahendiran, Gabriella Moraes, Mohith Shamdas, Christoph Kern, et al. A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis. *The Lancet Digital Health*, 1(6):e271–e297, 2019. doi: 10.1016/S2589-7500(19)30123-2.
- [12] Prasanna Porwal, Samiksha Pachade, Ravi Kamble, Manesh Kokare, Girish Deshmukh, Vivek Sahasrabuddhe, and Fabrice Meriaudeau. Indian diabetic retinopathy image dataset (IDRiD): A database for diabetic retinopathy screening research. *Data*, 3(3):25, 2018. doi: 10.3390/data3030025.
- [13] Prasanna Porwal, Samiksha Pachade, Manesh Kokare, Girish Deshmukh, Jaemin Son, Woong Bae, Lihong Liu, Jianzong Wang, Xinhui Liu, Liangxin Gao, et al. IDRiD: Diabetic retinopathy – segmentation and grading challenge. *Medical Image Analysis*, 59:101561, 2020. doi: 10.1016/j.media.2019.101561.
- [14] Zhe Wang, Yanxin Yin, Jianping Shi, Wei Fang, Hongsheng Li, and Xiaogang Wang. Zoom-in-net: Deep mining lesions for diabetic retinopathy detection. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2017*, volume 10435 of *Lecture Notes in Computer Science*, pages 267–275. Springer, 2017. doi: 10.1007/978-3-319-66179-7_31.
- [15] Rui Sun, Yihao Li, Tianzhu Zhang, Zhendong Mao, Feng Wu, and Yongdong Zhang. Lesion-aware transformers for diabetic retinopathy grading. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10938–10947, 2021.
- [16] Jacob Cohen. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220, 1968. doi: 10.1037/h0026256.

- [17] Zhenxing Niu, Mo Zhou, Le Wang, Xinbo Gao, and Gang Hua. Ordinal regression with multiple output CNN for age estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4920–4928, 2016.
- [18] Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140: 325–331, 2020. doi: 10.1016/j.patrec.2020.11.008.
- [19] Jordi de la Torre, Domenec Puig, and Aida Valls. Weighted kappa loss function for multi-class classification of ordinal data in deep learning. *Pattern Recognition Letters*, 105:144–154, 2018. doi: 10.1016/j.patrec.2017.05.018.
- [20] Sudhir Varma and Richard Simon. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 7:91, 2006. doi: 10.1186/1471-2105-7-91.
- [21] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.
- [22] Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991. doi: 10.1162/neco.1991.3.1.79.
- [23] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*, 2017.
- [24] Asia Pacific Tele-Ophthalmology Society. APTOS 2019 blindness detection. Kaggle competition, <https://www.kaggle.com/competitions/aptos2019-blindness-detection>, 2019.
- [25] Nikos Tsiknakis, Dimitris Theodoropoulos, Georgios Manikis, Emmanouil Ktistakis, Ourania Boutsora, Alexa Berto, Fabio Scarpa, Alberto Scarpa, Dimitrios I. Fotiadis, and Kostas Marias. Deep learning for diabetic retinopathy detection and classification based on fundus images: A review. *Computers in Biology and Medicine*, 135:104599, 2021. doi: 10.1016/j.combiomed.2021.104599.
- [26] Harry Pratt, Frans Coenen, Deborah M. Broadbent, Simon P. Harding, and Yalin Zheng. Convolutional neural networks for diabetic retinopathy. *Procedia Computer Science*, 90: 200–205, 2016. doi: 10.1016/j.procs.2016.07.014.
- [27] Michael David Abramoff, Yiyue Lou, Ali Erginay, Warren Clarida, Ryan Amelon, James C. Folk, and Meindert Niemeijer. Improved automated detection of diabetic retinopathy on a publicly available dataset through integration of deep learning. *Investigative Ophthalmology & Visual Science*, 57(13):5200–5206, 2016. doi: 10.1167/iovs.16-19964.
- [28] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [29] Mingxing Tan and Quoc V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 6105–6114, 2019.
- [30] Mingxing Tan and Quoc V. Le. EfficientNetV2: Smaller models and faster training. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 10096–10106, 2021.

- [31] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11976–11986, 2022.
- [32] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- [33] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10012–10022, 2021.
- [34] Yukun Zhou, Mark A. Chia, Siegfried K. Wagner, Murat S. Ayhan, Dominic J. Williamson, Robbert R. Struyven, Timing Liu, Moucheng Xu, Mateo G. Lozano, Peter Woodward-Court, et al. A foundation model for generalizable disease detection from retinal images. *Nature*, 622:156–163, 2023. doi: 10.1038/s41586-023-06555-x.
- [35] Etienne Decencière, Xiwei Zhang, Guy Cazuguel, Bruno Lay, Béatrice Cochener, Caroline Trone, Philippe Gain, Richard Ordonez, Pascale Massin, Ali Erginay, Béatrice Charton, and Jean-Claude Klein. Feedback on a publicly distributed image database: the Messidor database. *Image Analysis & Stereology*, 33(3):231–234, 2014. doi: 10.5566/ias.1155.
- [36] Tao Li, Yingqi Gao, Kai Wang, Song Guo, Hanruo Liu, and Hong Kang. Diagnostic assessment of deep learning algorithms for diabetic retinopathy screening. *Information Sciences*, 501:511–522, 2019. doi: 10.1016/j.ins.2019.06.011.
- [37] Teresa Araújo, Guilherme Aresta, Luís Mendonça, Susana Penas, Carolina Maia, Ângela Carneiro, Ana Maria Mendonça, and Aurélio Campilho. DR|GRADUATE: Uncertainty-aware deep learning-based diabetic retinopathy grading in eye fundus images. *Medical Image Analysis*, 63:101715, 2020. doi: 10.1016/j.media.2020.101715.
- [38] Gwenolé Quéllec, Katia Charrière, Yassine Boudi, Béatrice Cochener, and Mathieu Lamard. Deep image mining for diabetic retinopathy screening. *Medical Image Analysis*, 39:178–193, 2017. doi: 10.1016/j.media.2017.04.012.
- [39] Yi Zhou, Xiaodong He, Lei Huang, Li Liu, Fan Zhu, Shanshan Cui, and Ling Shao. Collaborative learning of semi-supervised segmentation and classification for medical images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2079–2088, 2019.
- [40] Xiaomeng Li, Xiaowei Hu, Lequan Yu, Lei Zhu, Chi-Wing Fu, and Pheng-Ann Heng. CANet: Cross-disease attention network for joint diabetic retinopathy and diabetic macular edema grading. *IEEE Transactions on Medical Imaging*, 39(5):1483–1493, 2020. doi: 10.1109/TMI.2019.2951844.
- [41] Along He, Tao Li, Ning Li, Kai Wang, and Huazhu Fu. CABNet: Category attention block for imbalanced diabetic retinopathy grading. *IEEE Transactions on Medical Imaging*, 40(1): 143–153, 2021. doi: 10.1109/TMI.2020.3023463.
- [42] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7132–7141, 2018.

- [43] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon. CBAM: Convolutional block attention module. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 3–19, 2018.
- [44] Xiaolong Wang, Ross Girshick, Abhinav Gupta, and Kaiming He. Non-local neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7794–7803, 2018.
- [45] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [46] Chun-Fu Richard Chen, Quanfu Fan, and Rameswar Panda. CrossViT: Cross-attention multi-scale vision transformer for image classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 357–366, 2021.
- [47] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241. Springer, 2015.
- [48] Lingyu Wu, Haiying Xia, Shuxiang Song, and Yang Lan. HEL-Net: Heterogeneous ensemble learning for comprehensive diabetic retinopathy multi-lesion segmentation via Mamba-UNet. *Image and Vision Computing*, 166:105879, 2026. doi: 10.1016/j.imavis.2025.105879.
- [49] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. arXiv preprint arXiv:2312.00752, 2023.
- [50] Eibe Frank and Mark Hall. A simple approach to ordinal classification. In *Machine Learning: ECML 2001*, volume 2167 of *Lecture Notes in Computer Science*, pages 145–156. Springer, 2001.
- [51] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.
- [52] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-balanced loss based on effective number of samples. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9268–9277, 2019.
- [53] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2818–2826, 2016.
- [54] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009.
- [55] Simon Kornblith, Jonathon Shlens, and Quoc V. Le. Do better ImageNet models transfer better? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2661–2671, 2019.
- [56] Tal Ridnik, Emanuel Ben-Baruch, Asaf Noy, and Lihi Zelnik-Manor. ImageNet-21K pre-training for the masses. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

- [57] Maithra Raghu, Chiyuan Zhang, Jon Kleinberg, and Samy Bengio. Transfusion: Understanding transfer learning for medical imaging. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- [58] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [59] Boris T. Polyak and Anatoli B. Juditsky. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4):838–855, 1992. doi: 10.1137/0330046.
- [60] Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI)*, 2018.
- [61] Guotai Wang, Wenqi Li, Michael Aertsen, Jan Deprest, Sébastien Ourselin, and Tom Vercauteren. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. *Neurocomputing*, 338:34–45, 2019. doi: 10.1016/j.neucom.2019.01.103.
- [62] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017.
- [63] Gary S. Collins, Karel G. M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B. Reitsma, Maarten van Smeden, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.
- [64] Ben Graham. Kaggle diabetic retinopathy detection competition report. Technical report, University of Warwick, 2015.
- [65] Karel Zuiderveld. Contrast limited adaptive histogram equalization. In Paul S. Heckbert, editor, *Graphics Gems IV*, pages 474–485. Academic Press, 1994.
- [66] Zhun Zhong, Liang Zheng, Guoliang Kang, Shaozi Li, and Yi Yang. Random erasing data augmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 13001–13008, 2020.
- [67] Ross Wightman. PyTorch image models. <https://github.com/huggingface/pytorch-image-models>, 2019.
- [68] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [69] Leslie N. Smith and Nicholay Topin. Super-convergence: very fast training of neural networks using large learning rates. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications*, volume 11006 of *Proceedings of SPIE*, page 1100612, 2019. doi: 10.1117/12.2520589.
- [70] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed precision training. In *International Conference on Learning Representations (ICLR)*, 2018.
- [71] W. J. Youden. Index for rating diagnostic tests. *Cancer*, 3(1):32–35, 1950.

- 809 [72] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan,
810 Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. PyTorch: An imperative
811 style, high-performance deep learning library. In *Advances in Neural Information Processing*
812 *Systems (NeurIPS)*, volume 32, 2019.
- 813 [73] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion,
814 Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al.
815 Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–
816 2830, 2011.
- 817 [74] Jacob Gildenblat and contributors. PyTorch library for CAM methods. [https://github.](https://github.com/jacobgil/pytorch-grad-cam)
818 [com/jacobgil/pytorch-grad-cam](https://github.com/jacobgil/pytorch-grad-cam), 2021.
- 819 [75] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman &
820 Hall, New York, 1993.
- 821 [76] Thomas G. Dietterich. Approximate statistical tests for comparing supervised classifi-
822 cation learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998. doi: 10.1162/
823 089976698300017197.
- 824 [77] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi
825 Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-
826 based localization. In *Proceedings of the IEEE International Conference on Computer Vision*
827 *(ICCV)*, pages 618–626, 2017.
- 828 [78] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been
829 Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*
830 *(NeurIPS)*, volume 31, 2018.
- 831 [79] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions
832 and use interpretable models instead. *Nature Machine Intelligence*, 1:206–215, 2019. doi:
833 10.1038/s42256-019-0048-x.
- 834 [80] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. Dissecting racial
835 bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453,
836 2019. doi: 10.1126/science.aax2342.